

COPPER MOUNTAIN, CO, USA

WMO: 722061

Lat: 39.4752N

Lon: 106.1520W

Elev: 3680

StdP: 64.25

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 03038

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -21.1	(c) -18.7	(d) -34.2	(e) 0.2	(f) -4.1	(g) -30.0	(h) 0.4	(i) -5.3	(j) 16.8	(k) -10.5	(l) 15.5	(m) -9.5	(n) 6.1	(o) 270	(p) 0.908

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.5	(c) 20.0	(d) 6.5	(e) 18.0	(f) 6.1	(g) 17.5	(h) 6.1	(i) 9.6	(j) 15.2	(k) 8.8	(l) 14.4	(m) 8.1	(n) 14.1	(o) 5.3	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
7.3	10.1	12.3	6.2	9.3	11.6	5.1	8.6	11.3	38.9	14.8	36.8	14.5	34.8	14.2	12.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	13.6	11.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-8.2	-7.8	-4.3	-1.9	3.0	9.9	13.7	11.7	8.5	1.7	-3.2	-8.5
	DBStd	8.95	4.85	4.74	4.87	4.59	4.46	4.41	2.51	2.76	3.98	5.13	4.98	5.24
	HDD10.0	3451	566	498	442	357	219	53	4	14	68	259	396	575
	HDD18.3	6231	824	731	701	607	476	252	145	205	294	517	646	833
	CDD10.0	263	0	0	0	0	2	51	117	68	24	1	0	0
	CDD18.3	1	0	0	0	0	0	1	0	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	1	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	5.9	6.1	6.8	6.5	6.7	5.9	5.9	4.7	4.8	5.2	6.1	5.9	5.8
(15) Precipitation	PrecAvg	455	31	31	38	47	35	28	56	62	42	31	30	25
	PrecMax	615	100	75	62	77	104	55	108	94	57	53	48	78
	PrecMin	316	6	11	14	15	8	7	13	23	25	9	12	0
	PrecStd	73	23	18	14	16	21	14	28	20	9	12	10	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	3.0	7.8	10.9	15.1	22.2	22.4	20.1	19.7	14.7	7.9	2.9
		MCWB	-3.7	-3.2	-1.2	1.0	2.6	6.0	7.8	6.8	4.7	3.0	-0.5	-4.5
	2%	DB	1.2	2.1	5.3	7.4	12.4	18.8	20.2	17.9	17.2	12.2	6.1	1.3
		MCWB	-4.3	-3.8	-2.4	-0.9	2.2	4.8	7.0	6.6	5.0	2.4	-1.9	-5.0
	5%	DB	-0.2	0.9	3.0	6.2	11.3	17.6	18.8	17.4	15.2	9.9	4.7	0.2
		MCWB	-4.6	-4.1	-3.2	-1.6	1.8	4.6	6.8	6.5	4.4	1.4	-2.4	-5.0
	10%	DB	-2.1	-1.2	2.3	4.2	8.8	16.1	17.7	16.0	14.8	7.6	2.7	-1.3
		MCWB	-5.7	-5.0	-3.5	-2.5	0.7	4.0	6.7	6.1	4.4	0.4	-3.2	-5.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.8	-1.1	0.1	1.6	4.3	8.0	10.7	10.5	8.0	5.5	0.8	-2.2
		MCDB	1.4	2.4	5.0	6.7	11.4	16.0	16.3	15.3	14.0	10.4	4.6	0.4
	2%	WB	-3.2	-2.8	-1.1	0.4	3.2	7.0	9.8	9.4	6.9	3.8	-0.3	-3.3
		MCDB	-0.1	0.0	4.2	5.1	10.3	15.1	15.6	14.6	13.6	9.3	3.6	-0.3
	5%	WB	-4.3	-3.9	-2.1	-0.3	2.4	6.2	9.2	8.6	6.1	2.6	-1.4	-4.4
		MCDB	-1.5	-0.8	2.9	4.1	8.9	14.4	14.9	13.7	13.0	7.8	3.1	-1.3
	10%	WB	-5.1	-4.8	-3.1	-1.2	1.7	5.5	8.6	7.9	5.4	1.5	-2.3	-5.1
		MCDB	-2.4	-1.6	1.4	3.5	7.6	13.7	14.4	13.5	11.9	6.5	2.3	-1.9
(35) Mean Daily Temperature Range	MDBR		6.4	6.4	7.1	7.1	7.1	8.1	8.5	8.0	8.4	7.2	6.2	6.2
	5% DB	MCDBR	6.9	7.0	7.5	7.5	8.5	8.8	9.2	8.6	9.0	8.9	7.1	6.0
		MCWBR	4.6	4.6	4.4	4.0	4.0	3.9	3.8	3.8	4.0	4.3	3.8	4.1
	5% WB	MCDBR	6.5	6.0	7.0	7.0	7.8	8.4	8.3	8.0	8.2	8.0	6.7	5.7
		MCWBR	5.0	4.5	4.6	4.6	4.0	4.0	4.1	4.1	4.0	4.0	4.2	4.7
(40) Clear-Sky Solar Irradiance	taub		0.187	0.192	0.211	0.234	0.246	0.252	0.274	0.271	0.240	0.221	0.198	0.188
	taud		2.532	2.493	2.469	2.530	2.649	2.656	2.633	2.613	2.734	2.743	2.687	2.565
	Ebn at Noon		1042	1070	1068	1049	1034	1022	996	997	1022	1025	1024	1019
	Edh at Noon		70	85	98	100	91	91	92	91	75	66	60	62
(44) All-Sky Solar Radiation	RadAvg		2.72	3.73	5.08	5.75	6.13	7.20	6.41	5.66	5.13	3.81	2.72	2.25
	RadStd		0.12	0.19	0.38	0.40	0.48	0.43	0.34	0.29	0.35	0.29	0.22	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+69	+21

Nomenclature: See separate page